

Your Cancer Summary Report

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Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma with BRCA2
Germline Mutation

Precision Oncology Center — Demonstration Report

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⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Breast Cancer - ER+/PR+/HER2- Invasive Ductal Carcinoma

Stage: Stage IIA | **Grade:** Grade 2

You have Stage IIA breast cancer that is fuelled by estrogen and progesterone. This is an early-to-intermediate stage with strong treatment options. Genomic testing found two additional drug targets — BRCA2 and PIK3CA — opening further therapy choices.

✓ Positive Factors

- ER+/PR+ status — hormone-blocking therapy (tamoxifen) is active
- BRCA2 germline mutation — eligible for FDA-approved PARP inhibitor (olaparib)
- PIK3CA H1047R — targetable with alpelisib + fulvestrant (FDA-approved, SOLAR-1)
- CCND1 amp + CDKN2A del — both support CDK4/6 inhibitor benefit
- ESR1 Moran's I 0.42 — hormone-driven spatial clustering confirmed
- All 5 key breast cancer genes: FDR $q < 7 \times 10^{-11}$ across RNA, protein, phospho

⚠ Challenges to Consider

- BRCA2 germline — 50% transmission risk; family genetic counseling essential
- MYC amplification may contribute to future treatment resistance

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. Requires oncologist review before any treatment decisions.

- Immune-cold TME — tumour has partially excluded immune cells

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA2

Pathogenic

c.5946delT frameshift, germline (AF=0.50, COSV57492880)

You were born with a change in the BRCA2 DNA-repair gene. This raised your cancer risk but makes the tumour sensitive to PARP inhibitor drugs that force cancer cells to self-destruct.

Treatment Option: Olaparib / Niraparib — FDA-approved for germline BRCA1/2-mutated HER2- breast cancer

PIK3CA

Pathogenic

H1047R missense, somatic (AF=0.42, COSM775)

An activating change in PIK3CA is turbo-charging a growth signal in your cancer cells. An FDA-approved drug combination targets this directly.

Treatment Option: Alpelisib + fulvestrant — FDA-approved (SOLAR-1) for PIK3CA-mutated HR+/HER2- breast cancer

GATA3

Likely Pathogenic

Frameshift, somatic (AF=0.38, COSV104733902)

A GATA3 change confirms the hormone-driven identity of your cancer and is commonly seen in ER+ breast cancer.

Treatment Option: No direct targeted therapy; reinforces continued endocrine strategy

MYC

Pathogenic — Copy Number Gain

Amplification chr8q24 ($\log_2=1.45$, ~5 copies)

Extra MYC copies are driving rapid cell growth. Investigational BET inhibitor drugs are being studied against this target.

Treatment Option: BET inhibitors (investigational) — consider clinical trial

CCND1

Pathogenic — Copy Number Gain

Amplification chr11q13 ($\log_2=1.62$, ~6 copies)

Extra CCND1 copies accelerate cell division — exactly the target of CDK4/6 inhibitors, which are standard-of-care additions to hormone therapy.

Treatment Option: Palbociclib / Ribociclib / Abemaciclib — FDA-approved for ER+/HER2- breast cancer

CDKN2A

Pathogenic — Copy Number Loss

Deletion chr9p21.3 ($\log_2=-1.65$, ~1 copy)

A missing copy of the CDKN2A braking gene means the cell-division accelerator runs unchecked — CDK4/6 inhibitors are designed to compensate for exactly this.

Treatment Option: Supports CDK4/6 inhibitor use; CDKN2A loss may predict enhanced response

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

900-spot Visium analysis across 7 tissue regions. ESR1 mean 544 counts in tumour_core_luminal vs 17 in stroma (32-fold). CD8A/CD68 concentrated in stroma_immune — immune-excluded pattern. ESR1 Moran's I = 0.42 (significantly clustered, $z=264.7$). MKI67 Moran's I = 0.25 (heterogeneous proliferation).

Immune Status: • Low immune activity - The tumor may be evading your immune system

Your tumour is strongly driven by oestrogen signalling concentrated in its centre. Immune cells are largely excluded from the tumour core, sitting around the edges instead. This confirms hormone-blocking therapy is hitting the right target.

Key Patterns Observed

- ESR1 32-fold enriched in tumour core vs stroma — dominant luminal ER biology
- PGR (511) and GATA3 (665) co-localise with ESR1 in tumour core
- CD8A highest in stroma_immune (203) — immune-excluded pattern
- MKI67 spatially heterogeneous (Moran's I 0.25) — uneven proliferation

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen + CDK4/6 Inhibitor (palbociclib / ribociclib / abemaciclib)

Endocrine +
targeted

FDA Approved

Expected Response: 60-70% PFS improvement vs. endocrine alone

Your current tamoxifen blocks oestrogen. Adding a CDK4/6 inhibitor (pills in cycles) further slows division. CCND1 amp and CDKN2A del in your tumour predict strong benefit.

Why this may help you: ER+/PR+ biology plus CCND1 amplification and CDKN2A deletion — textbook CDK4/6 indication

Common side effects: Neutropenia — monthly CBC monitoring required, Fatigue, Nausea, Hot flashes

Olaparib (PARP Inhibitor)

Targeted — PARP inhibitor

FDA Approved

Expected Response: 57-60% ORR (OlympiAD trial)

Olaparib exploits the BRCA2 repair defect — cancer cells accumulate fatal DNA damage while normal cells cope.

Why this may help you: Pathogenic germline BRCA2 frameshift — FDA-approved indication

Common side effects: Nausea, Fatigue, Anaemia, Decreased appetite

Alpelisib + Fulvestrant

Targeted — PI3Ka inhibitor + endocrine

FDA Approved

Expected Response: 35-40% PFS benefit (SOLAR-1)

Alpelisib blocks the PIK3CA H1047R growth signal directly. Combined with fulvestrant it attacks the cancer at two levels at once.

Why this may help you: PIK3CA H1047R at 42% allele frequency — the exact mutation in the SOLAR-1 FDA approval

Common side effects: Hyperglycaemia — requires close blood-sugar monitoring, Rash, Diarrhoea, Nausea

Clinical Trials

You may be eligible for these research studies offering new treatments:

OlympiA: Adjuvant Olaparib in Germline BRCA-Mutated HER2-Negative Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT03796040

Adjuvant olaparib for germline BRCA2, HER2-negative breast cancer — your exact profile. Ask your oncologist if staging makes you eligible.

Contact: clinicaltrials.gov/ct2/show/NCT03796040

INAVO120: Inavolisib + Palbociclib + Fulvestrant in PIK3CA-Mutated HR+ /HER2- Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT05242367

Next-generation PI3K inhibitor targeting your PIK3CA H1047R mutation combined with CDK4/6 inhibitor and fulvestrant.

Contact: clinicaltrials.gov/ct2/show/NCT05242367

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 6 months Breast MRI or mammogram
Monitor tumour response and detect new lesions

Every 3 months CEA and CA 15-3
Current levels normal; rising values signal possible progression

Monthly on CDK4/6 or PARP inhibitor CBC (complete blood count)
Detect bone-marrow suppression

Monthly if starting alpelisib HbA1c / blood glucose
Alpelisib commonly causes high blood sugar

Annually Pelvic ultrasound + gynaecologic exam
BRCA2 germline increases ovarian cancer risk

Within 30 days Genetic counselling referral
50% transmission risk to first-degree relatives

When to Call Your Doctor

Contact your care team right away if you experience:

- New or worsening bone pain
- Shortness of breath or chest pain
- Fever above 38°C on CDK4/6 inhibitor
- Unusual bruising or bleeding
- Extreme fatigue or pallor

Questions? Oncology nurse navigator or 24-hour triage line

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For Your Family

The BRCA2 mutation is inherited and present in every cell of your body. Each child has a 50% chance of inheriting it. Siblings and parents also have a 50% chance. BRCA2 carriers face a 45-85% lifetime breast cancer risk and 10-20% ovarian cancer risk.

Recommended Steps for Family Members

- Schedule genetic counselling within 4 weeks
- Inform first-degree relatives and recommend testing
- Discuss risk-reducing surgery with your gynaecologist
- Consider referral to a hereditary cancer programme

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:



BRCA Foundation

Education, support, and genetic testing resources for BRCA1/2 carriers and families

📞 1-800-292-2776 | 🌐 <https://brcafoundation.org>



Susan G. Komen Helpline

Breast cancer helpline, financial assistance, and peer community support

📞 1-877-465-6636 | 🌐 <https://www.komen.org>



FORCE — Facing Our Risk of Cancer Empowered

Dedicated support for hereditary breast and ovarian cancer; BRCA-specific peer navigation

| 🌐 <https://www.facingourrisk.org>

⚠️ Important Notice

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Data Sources: mockepic (DRY_RUN), fgbio (DRY_RUN), genomic-results (fixture files), spatialtools (Visium fixture data), multiomics (Stouffer meta)

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